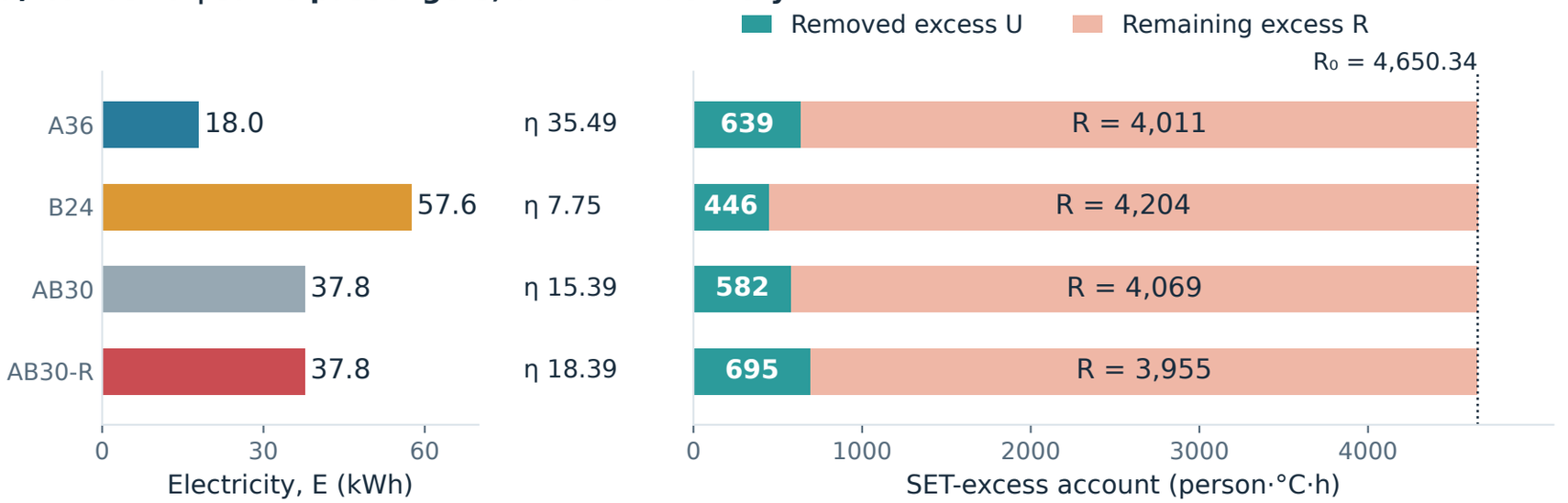
From electrical input to useful service

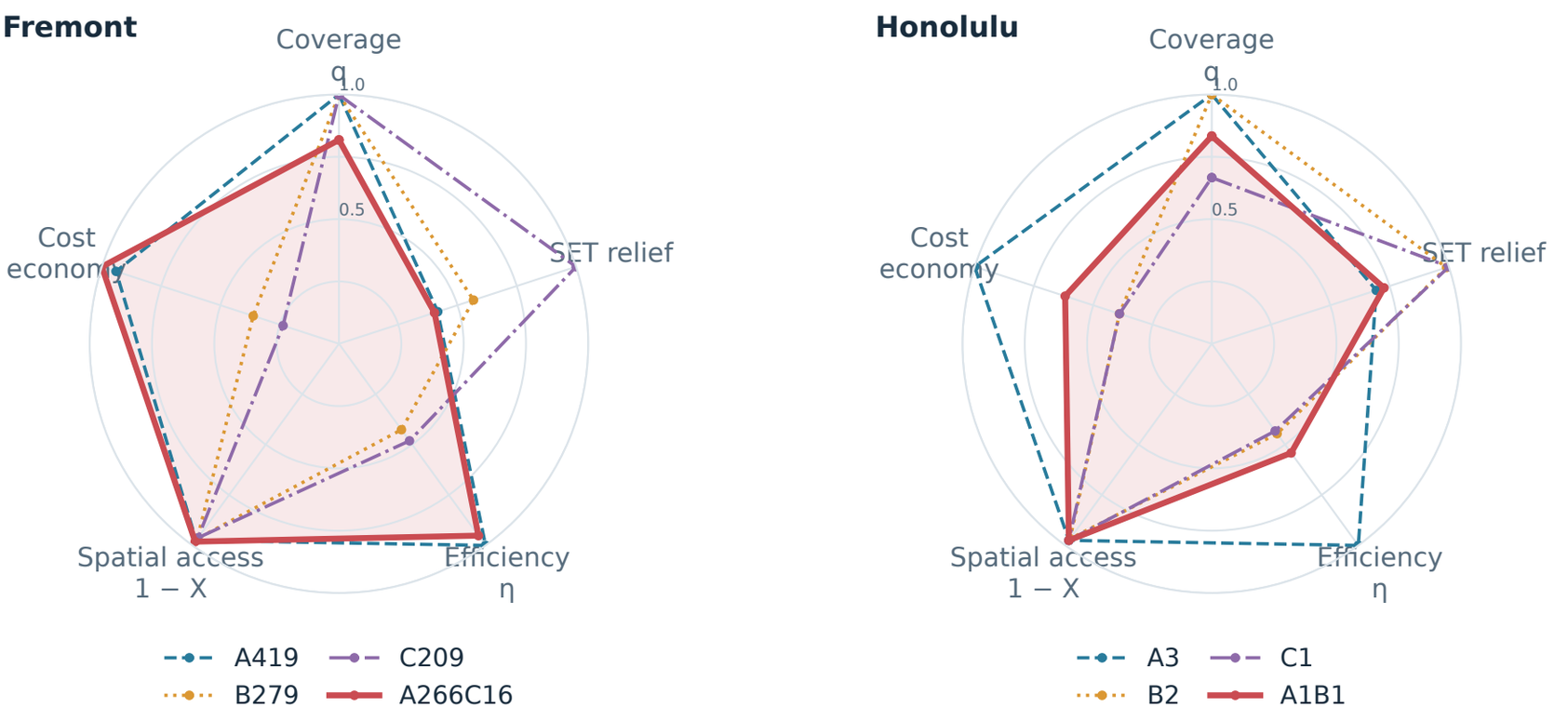
Separate energy and discomfort accounts; compare operational trade-offs without mixing units

(a) Canberra | same passengers, different delivery



At 37.8 kWh, redistribution delivers +19.5% useful service. U + R = R₀.

(b) Venue trade-offs | larger radius is better on each stated scale



q and 1 − X use their absolute 0–1 scale. Relief and η: fraction of the maximum shown within each venue.
Cost economy: minimum shown annual cost / plan cost. This diagnostic adds no optimization objective.